NANYANG TECHNOLOGICAL UNIVERSITY

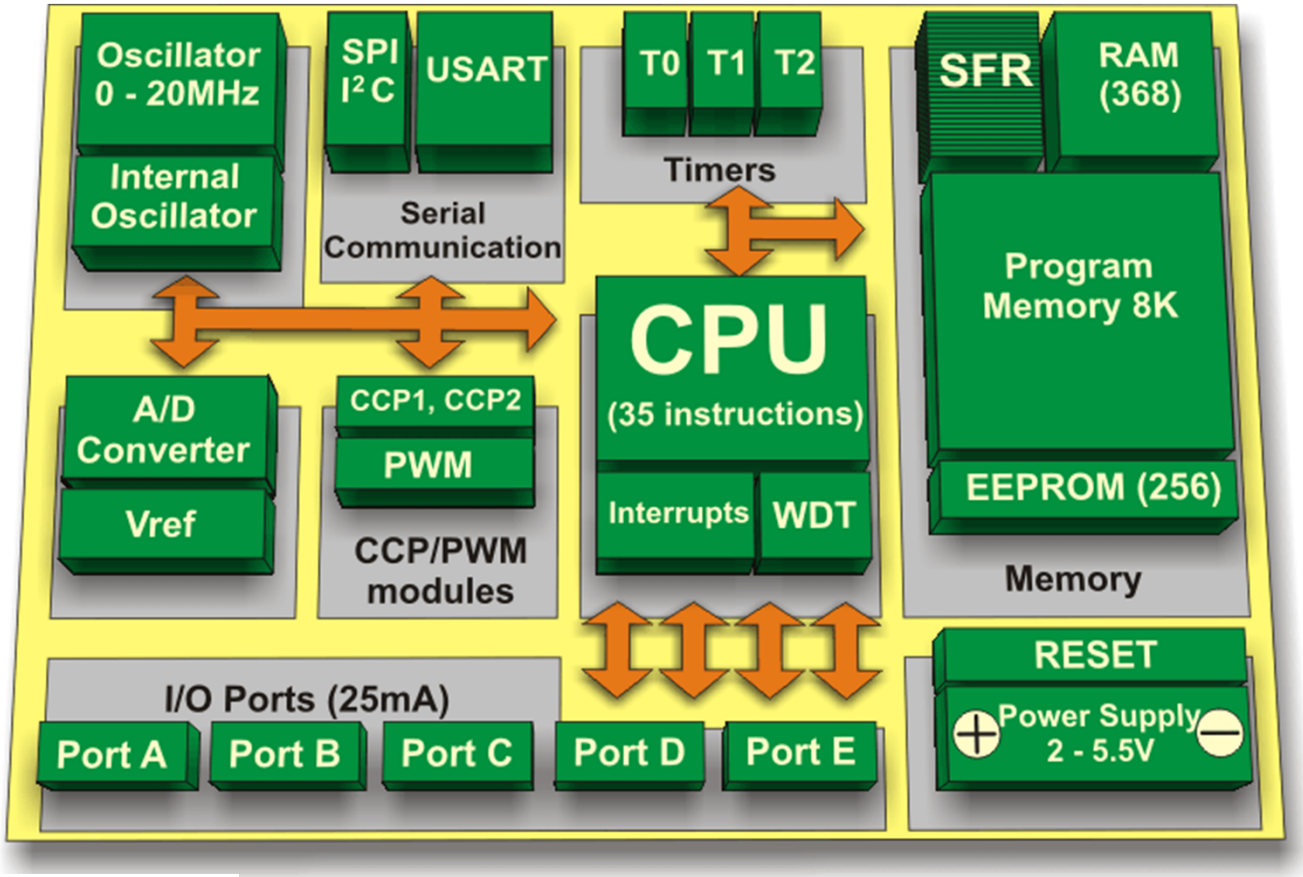
SC2107/CE2107
Microprocessor System Design and Development

Display

I/O Devices for Embedded Systems

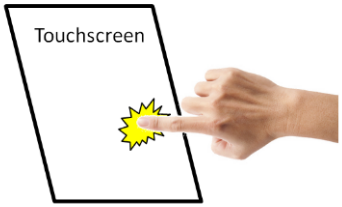
NANYANG TECHNOLOGICAL UNIVERSITY

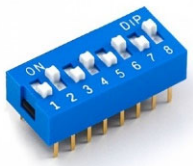
Microcontroller





Pushbutton switch





Switches



LCD



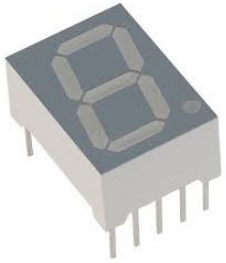
Keypad



LEDs



Dot Matrix LED



7 segment display

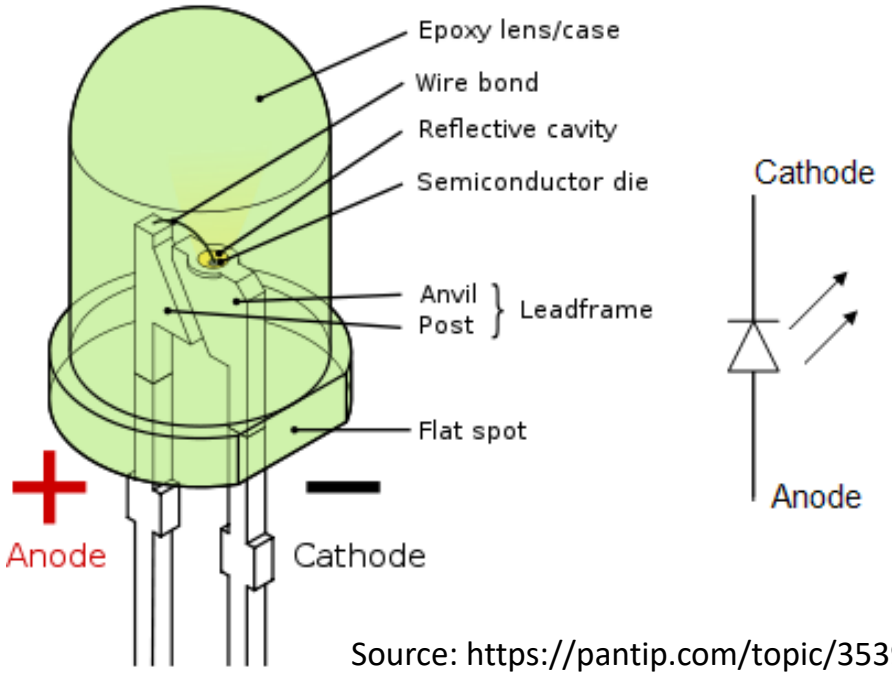
Topics Covered

- Light Emitting Diode (LED)
- Seven-Segment Display
- Dot Matrix LED
- Liquid Crystal Display (LCD)

Light Emitting Diode (LED)

Light Emitting Diode (LED)

- LED is a semiconductor light source.
- Illuminated by the movement of electrons in a semiconductor material.
- Different LED compounds produce different intensity levels.



Source: https://pantip.com/topic/35396323

Typical LED Characteristics			
Semiconductor Material	Wavelength	Colour	V_F @ 20mA
GaAs	850-940nm	Infra-Red	1.2v
GaAsP	630-660nm	Red	1.8v
GaAsP	605-620nm	Amber	2.0v
GaAsP:N	585-595nm	Yellow	2.2v
AlGaP	550-570nm	Green	3.5v
SiC	430-505nm	Blue	3.6v
GaInN	450nm	White	4.0v

Source: http://www.electronics-tutorials.ws/diode/diode_8.html

Light Emitting Diode (LED)

NANYANG TECHNOLOGICAL UNIVERSITY

- Advantages of LEDs:
  - Small size.
  - Low cost.
  - Low power consumption.
  - Reliable - has long lifespan – 10K+ hours



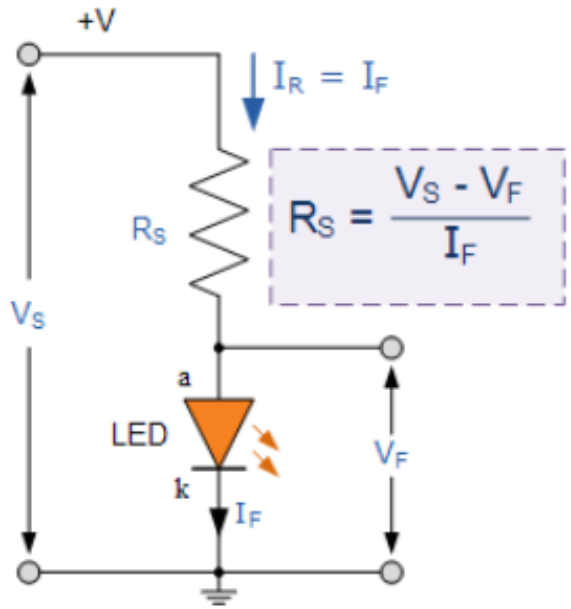
VS.



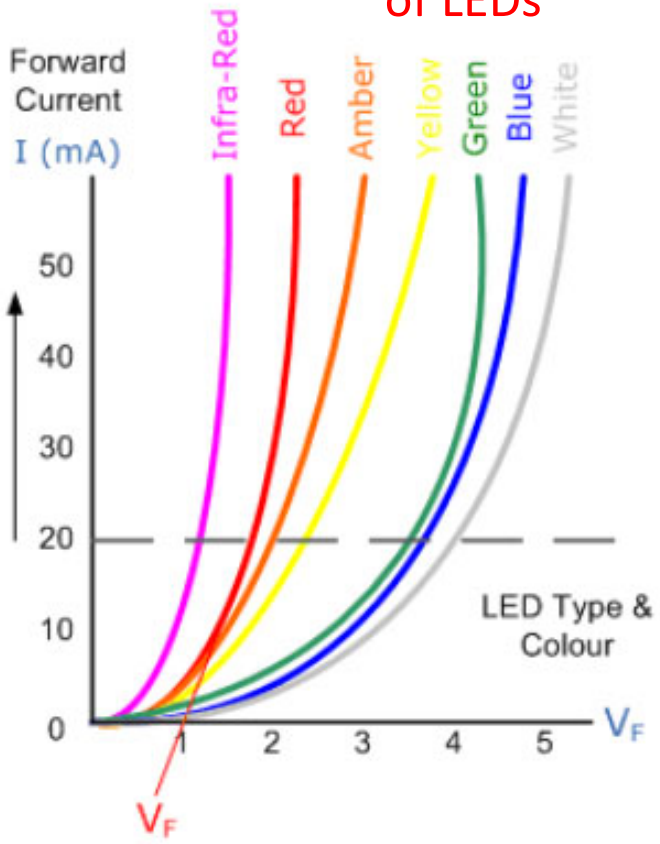
Source: http://sew-lexicon.com/

LED Interfacing

- When interfacing with a microcontroller pin, LED should be current limited using a series resistor to protect it from excessive current flow.
- The resistor value R_S is calculated using Ohm's Law given the required forward current I_F and forward voltage drop of the LED V_F, and the supply voltage V_S.



IV Characteristics of LEDs

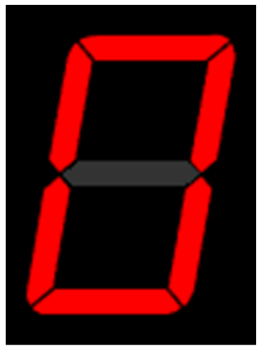


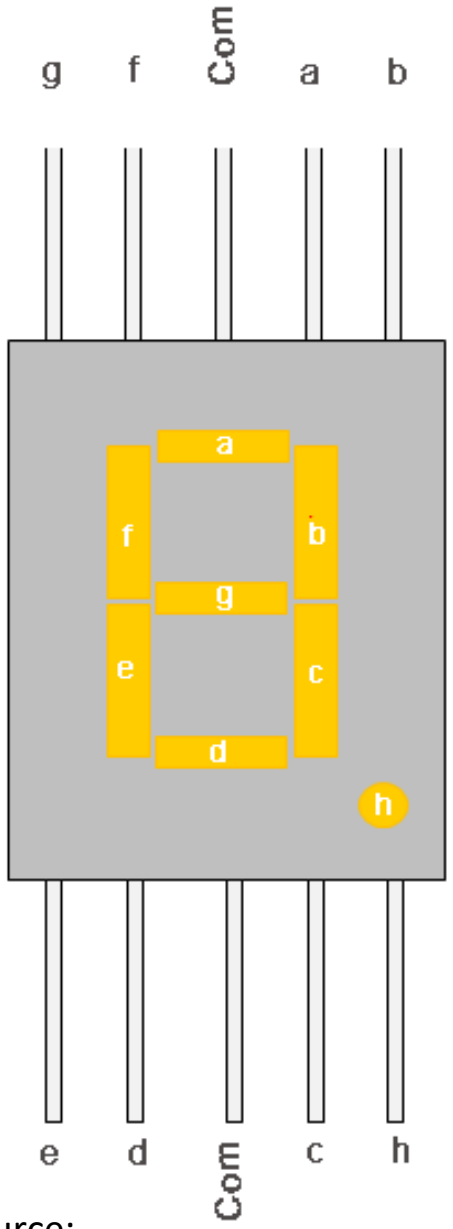
Source: http://www.electronics-tutorials.ws/diode/diode_8.html

Seven-Segment Display

Seven-Segment Display

- Consists of seven LED segments arranged in a rectangular fashion and one Decimal Point.
- When illuminated, each segment forms part of a numerical digit (both Decimal and Hex) to be displayed.
- Widely used in digital clocks, electronic meters and low-cost devices.
- Two configurations:
  - Common Cathode (CC)
  - Common Anode (CA)

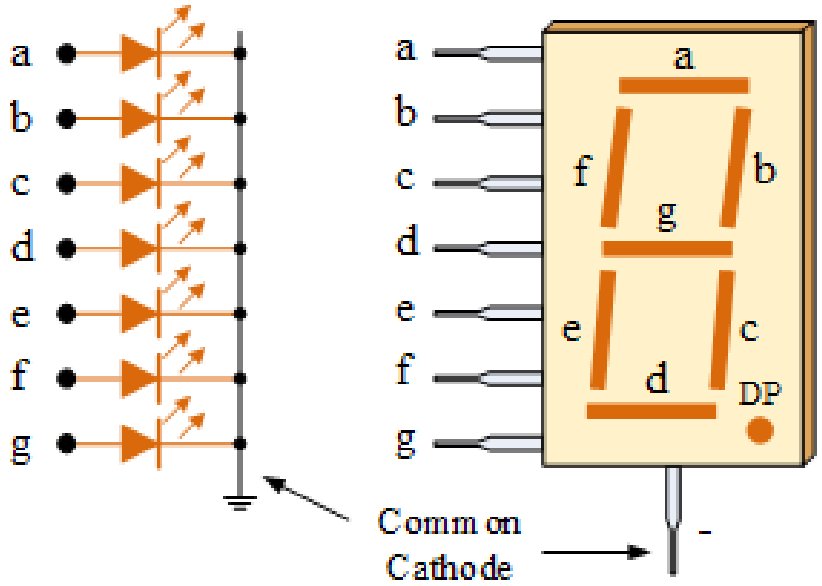




Source: https://circuitdigest.com/article/7-segment-display

Common Cathode (CC)

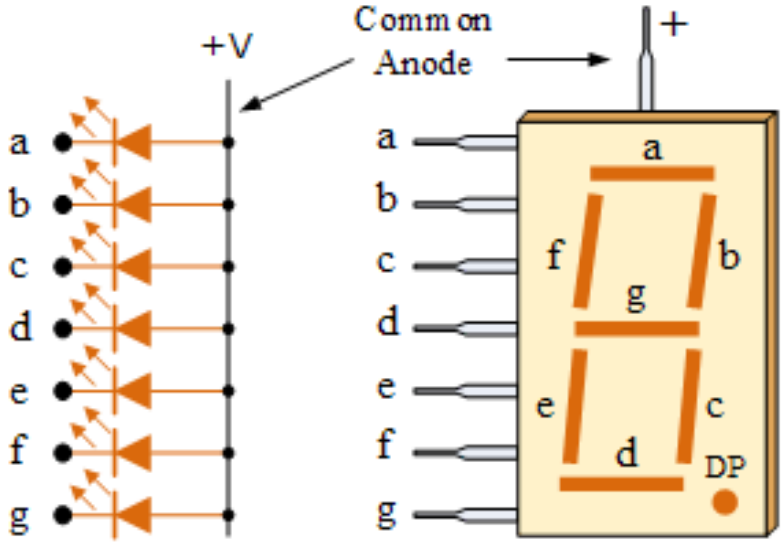
- All the cathode connections of the LED segments are joined together to logic “0” or ground.
- Individual segments are illuminated by applying a logic “1” signal via a current limiting resistor to forward bias the individual Anode terminals (a-g).



Source: http://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html

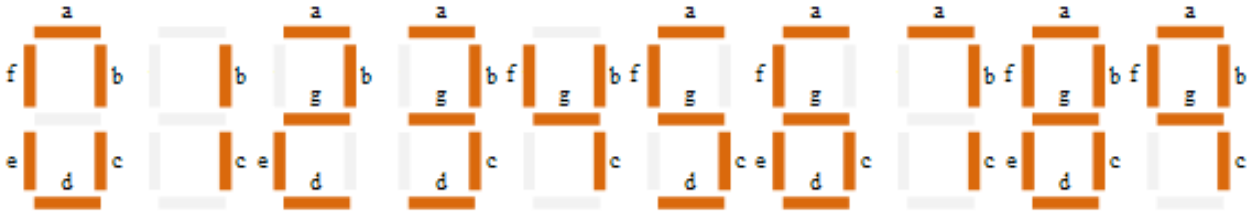
Common Anode (CA)

- All the anode connections of the LED segments are joined together to logic “1”.
- Individual segments are illuminated by applying a logic “0” via a suitable current limiting resistor to the Cathode of the particular segment (a-g).
- Generally more popular.



Source: http://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html

Display Numbers on 7-Segment Display

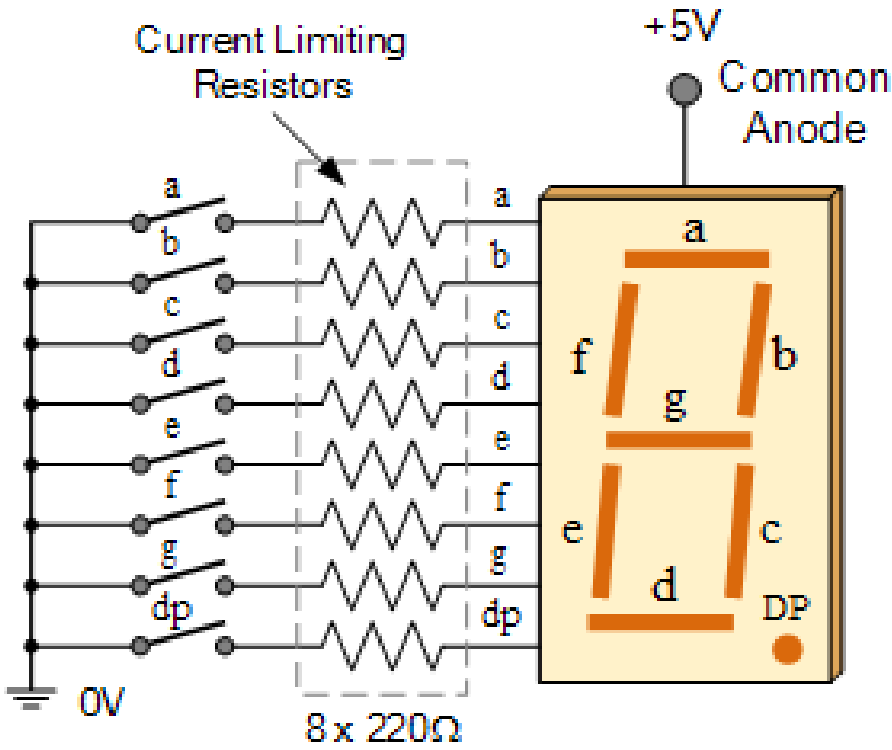


Truth table for displaying digits on Common Anode type 7-Segment Display

Digit to Display	h g f e d c b a	Hex code
0	11000000	C0
1	11111001	F9
2	10100100	A4
3	10110000	B0
4	10011001	99
5	10010010	92
6	10000010	82
7	11111000	F8
8	10000000	80
9	10010000	90

Interfacing to 7-Segment Display

- Each of the seven individual LEDs within a single package needs protection from over current by using current limiting registers.
- Number of switches can be reduced by using a BCD to 7-segment decoder.

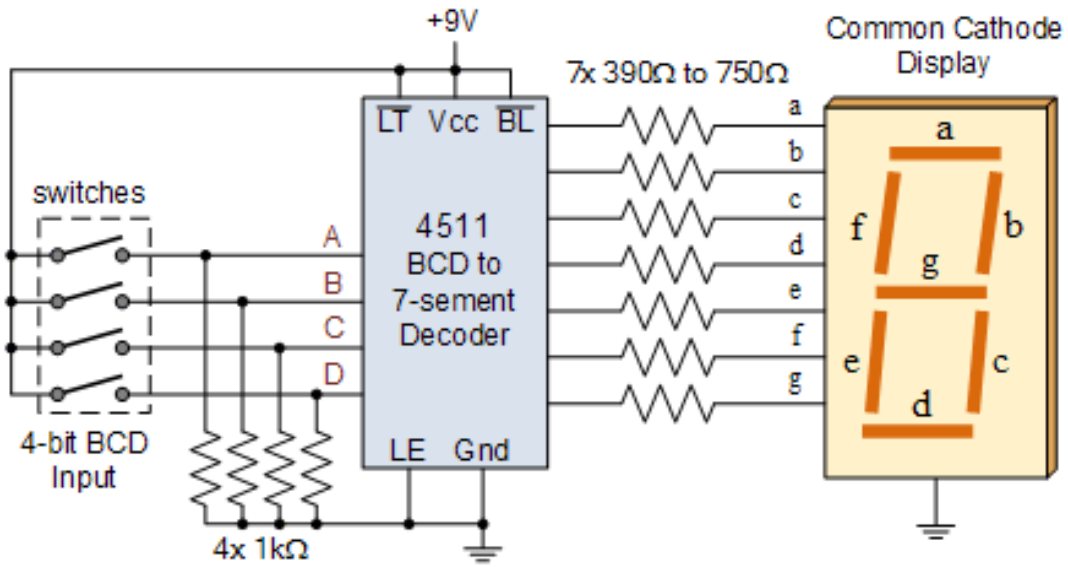


Source: http://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html

Interfacing to 7-Segment Display

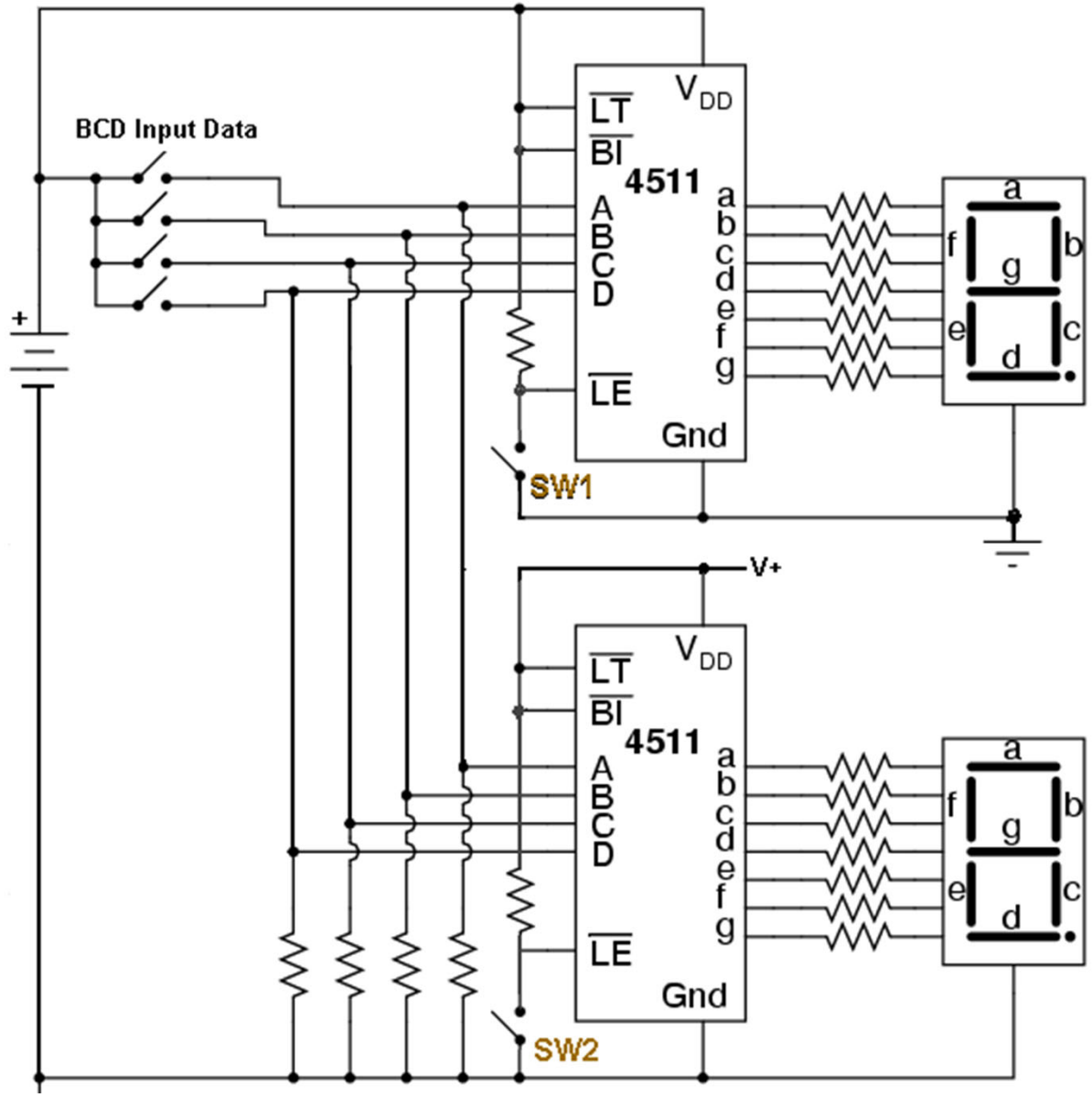
- Using the BCD to 7-segment decoder, the 4 toggle switches are used to select the desired numeral (0-9) for display.
- LT (Lamp Test) turns all segments ON so we can verify that all the segments are working
- Blanking (BL) forces the entire display OFF.
- LE is used to control the latching in of new data into the decoder.

Source: http://www.electronics-tutorials.ws/blog/7-segment-display-tutorial.html



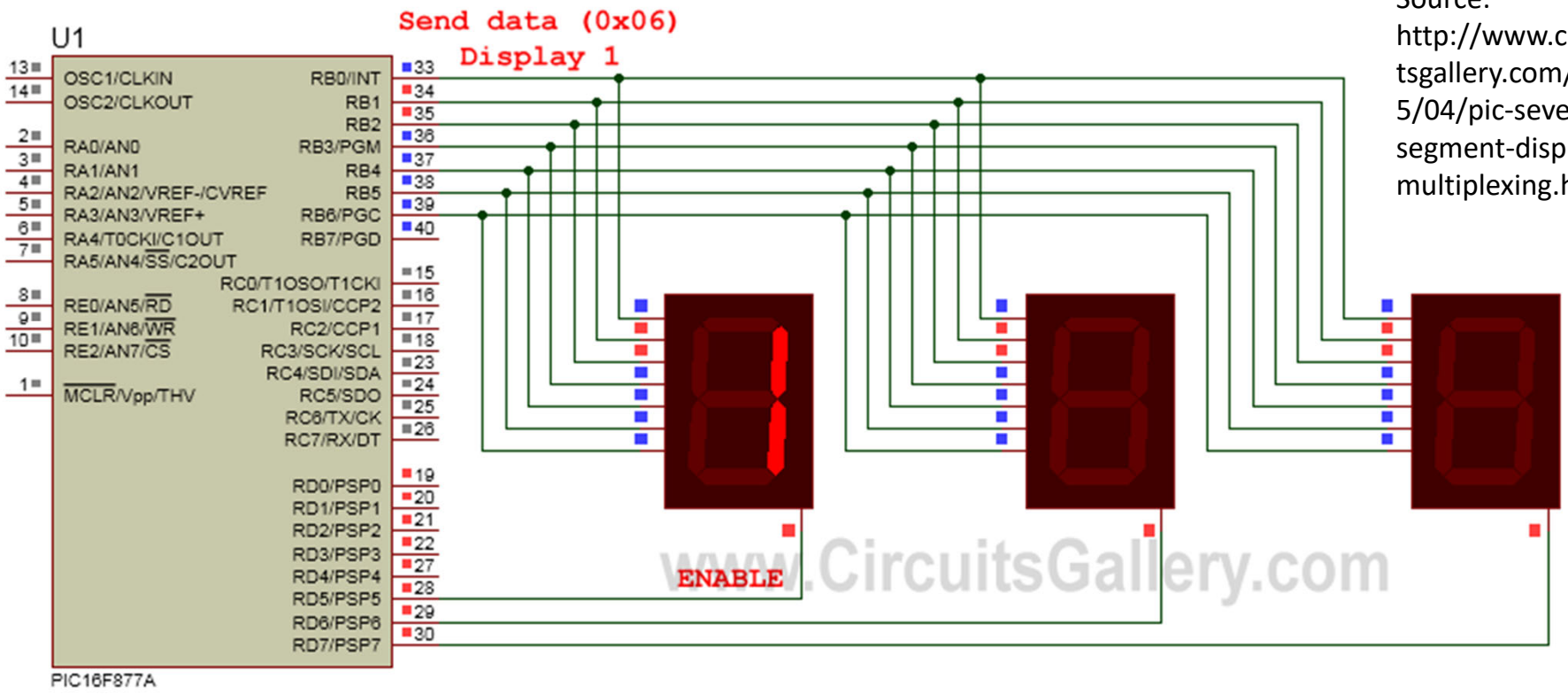
Interfacing to 7-Segment Display

- Multiple decoders can be used to control several 7-segment displays.
- By controlling the state of the LE pin, we can use the same input data lines to selectively write to each of the 7-segment display independently.



Multiplexed 7-Segment Display

- Human eye cannot notice that visual change if rate is 25 (or more) frames per second.
- To display all 3 digits, each 7-segment display is activated sequentially using an appropriate frequency so that it will appear as if they are ALL ON at the same time.



Source: http://www.circuitsgallery.com/2015/04/pic-seven-segment-display-multiplexing.html

Multiplexed 7-Segment Display

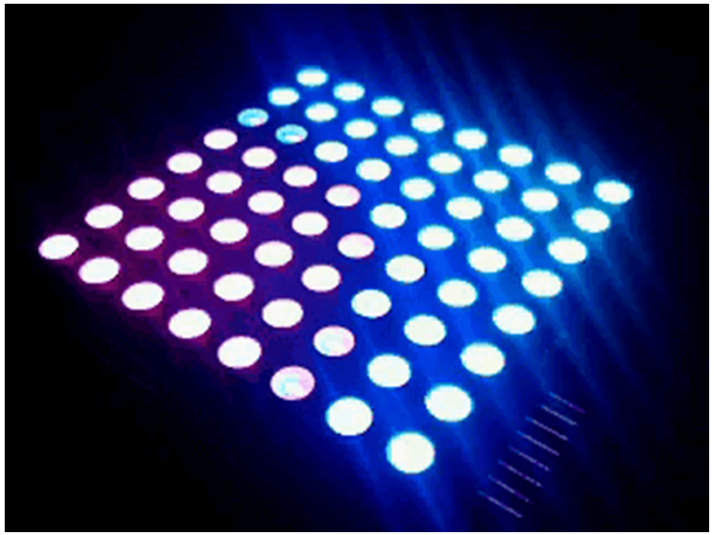
- Multiplexing can be achieved by directly controlling the CA or CC pin.
  - One ENABLE pin at each 7-segment display (Common Anode or Common Cathode).
- All the A, B, C, D, E, F, G, DP pins of 3 units are shorted to each other and connected to the microcontroller.
- Each 7-segment display is sequentially activated by sending corresponding data and enable signal at a frequency of more than 25Hz.

Dot Matrix LED

Dot Matrix LED

- Dot matrix LED display contains the group of LEDs as a two dimensional array.
- Enables display of both static and animated text and images.

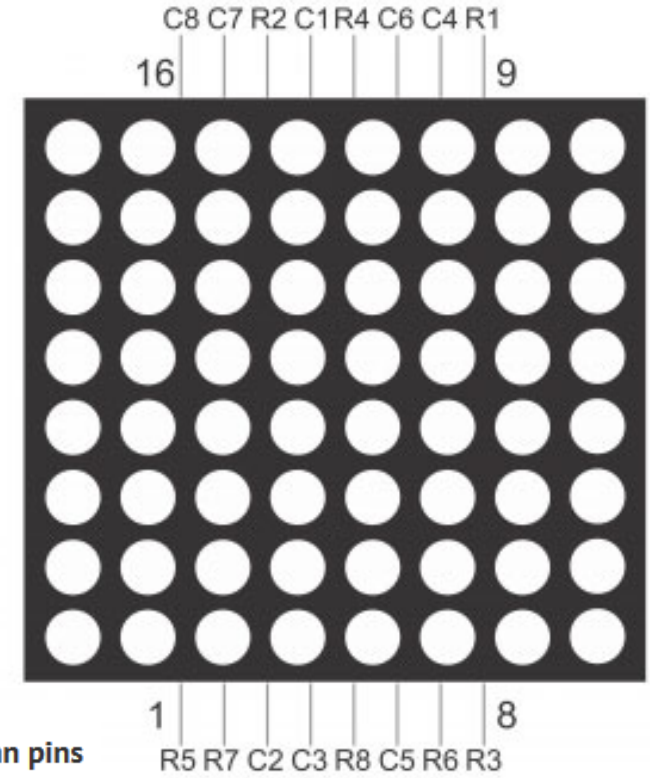




Source: https://www.seeedstudio.com/8x8-RGB-LED-Dot-Matrix-Compatible-with-Rainbowduino-p-113.html

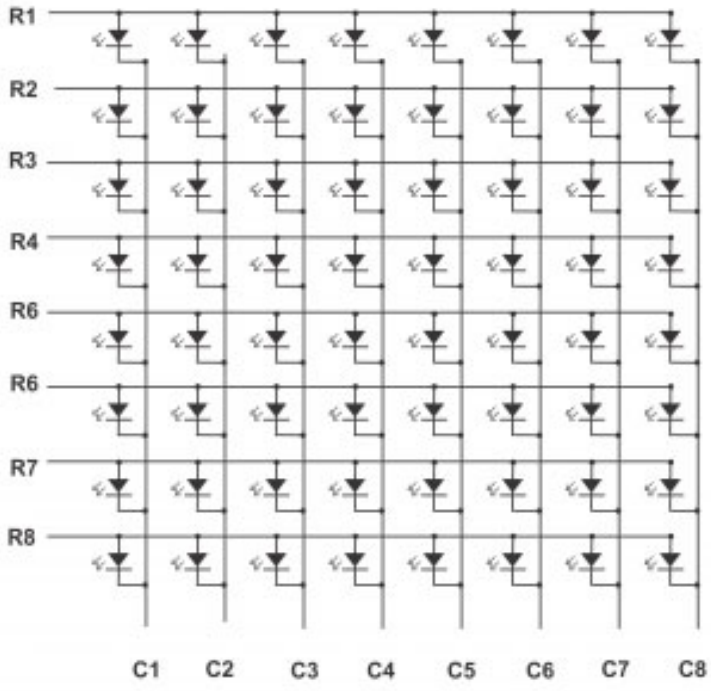
Dot Matrix LED

- The anode of each LED are connected together in the same column and the cathode of each LED are connected together in the same row (or vice versa).
- For example, 8×8 LED matrix requires only 16 I/O pins (instead of 64).



C1 – C8 – Column pins

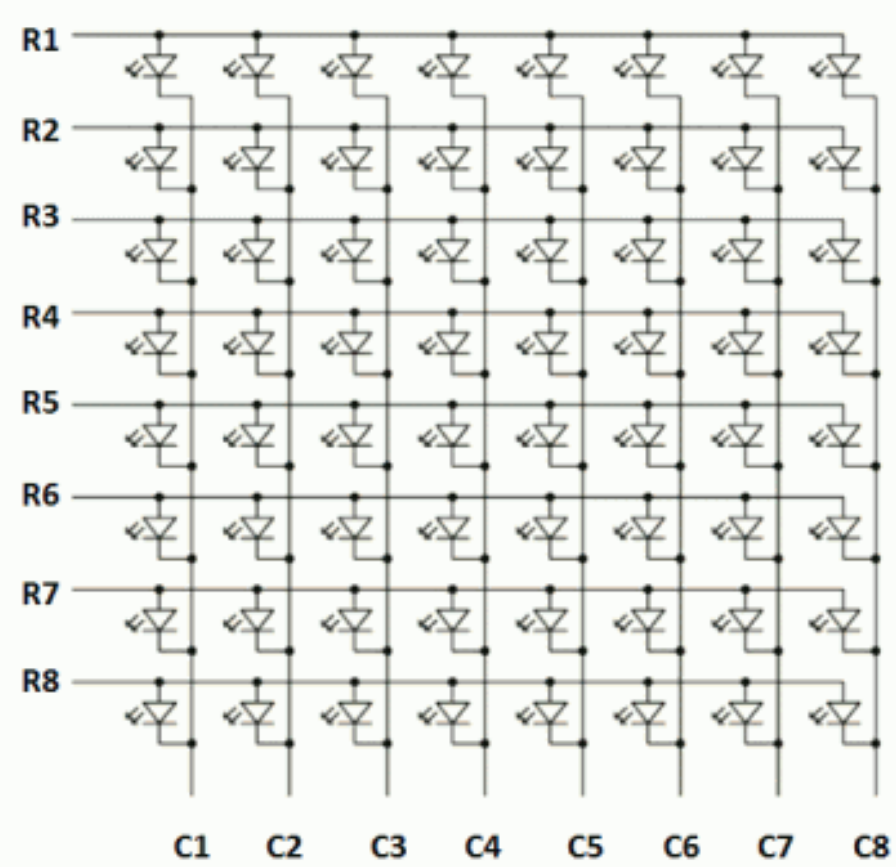
R1 – R8 – Row pins



Source: http://www.circuitstoday.com/interfacing-8x8-led-matrix-with-arduino

Dot Matrix LED

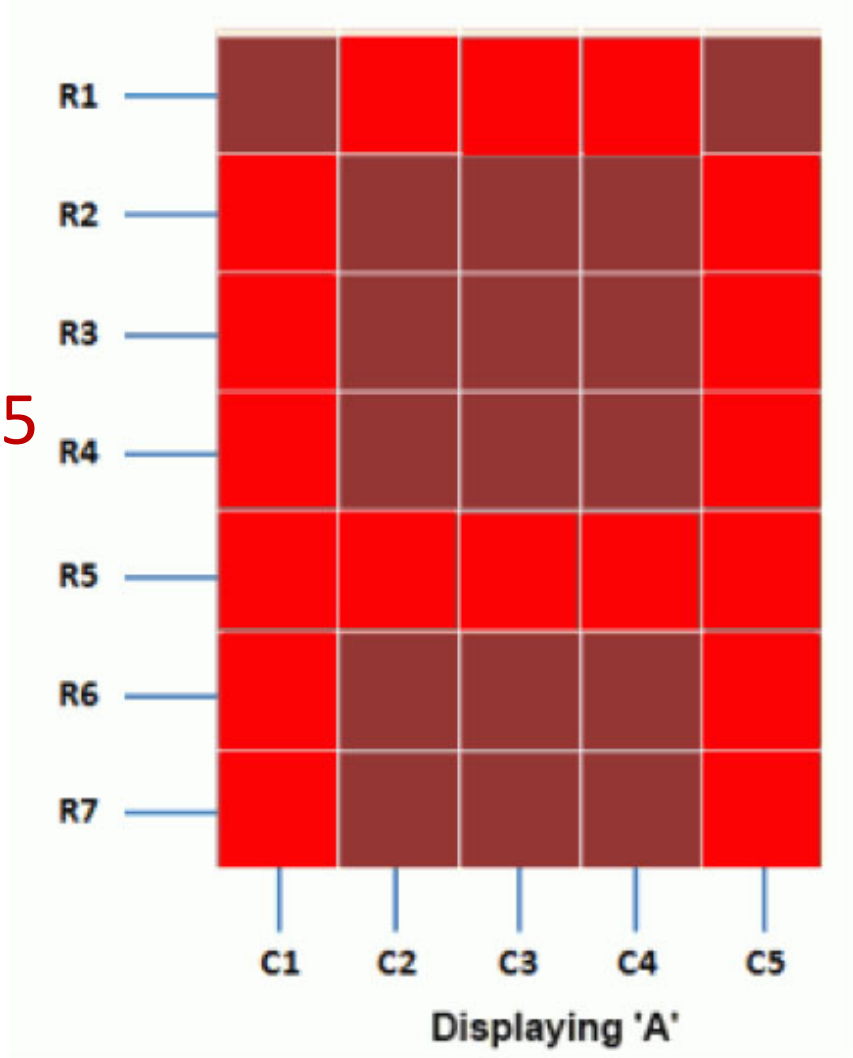
- Each LED is addressed by its row and column number.
- If R4 is pulled high and C3 is pulled low, the LED in 4th row and 3rd column will be turned ON.
- Characters can be displayed by fast scanning of rows or columns.
- We will discuss the Column Scanning approach.



Source: http://embedded-lab.com/blog/lab-12-basics-of-led-dot-matrix-display/

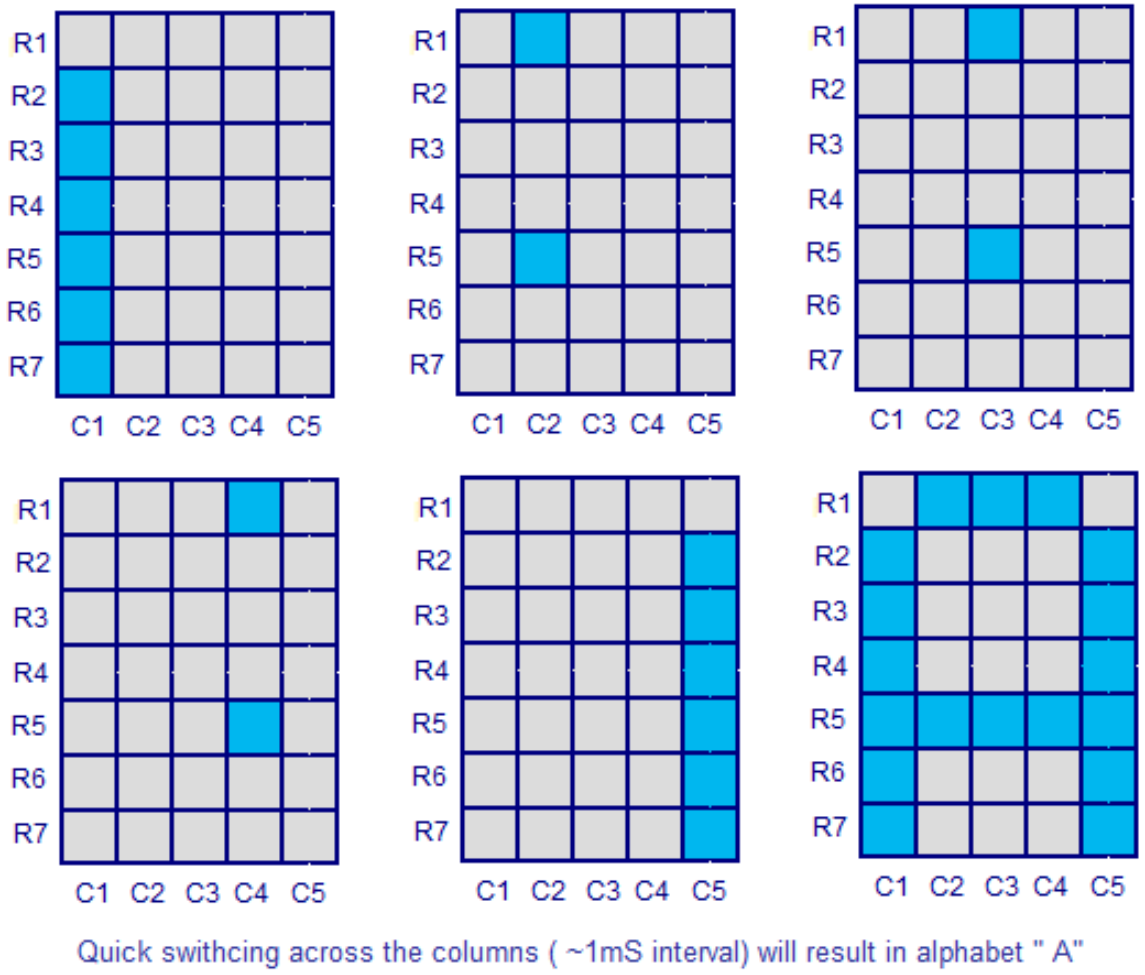
Displaying ‘A’ on 5x7 matrix of LEDs

- Scan across the columns quickly (> 100 times per second), and turn on the respective LEDs in each row of that column
  - Select C1 (C1 is pulled low) and deselect other columns (C2 to C5 are pulled high).
  - Select R2 to R7 (Apply forward bias).
  - Next select C2 and deselect the other columns.
  - Select R1 and R5.
  - Proceed with remaining columns.



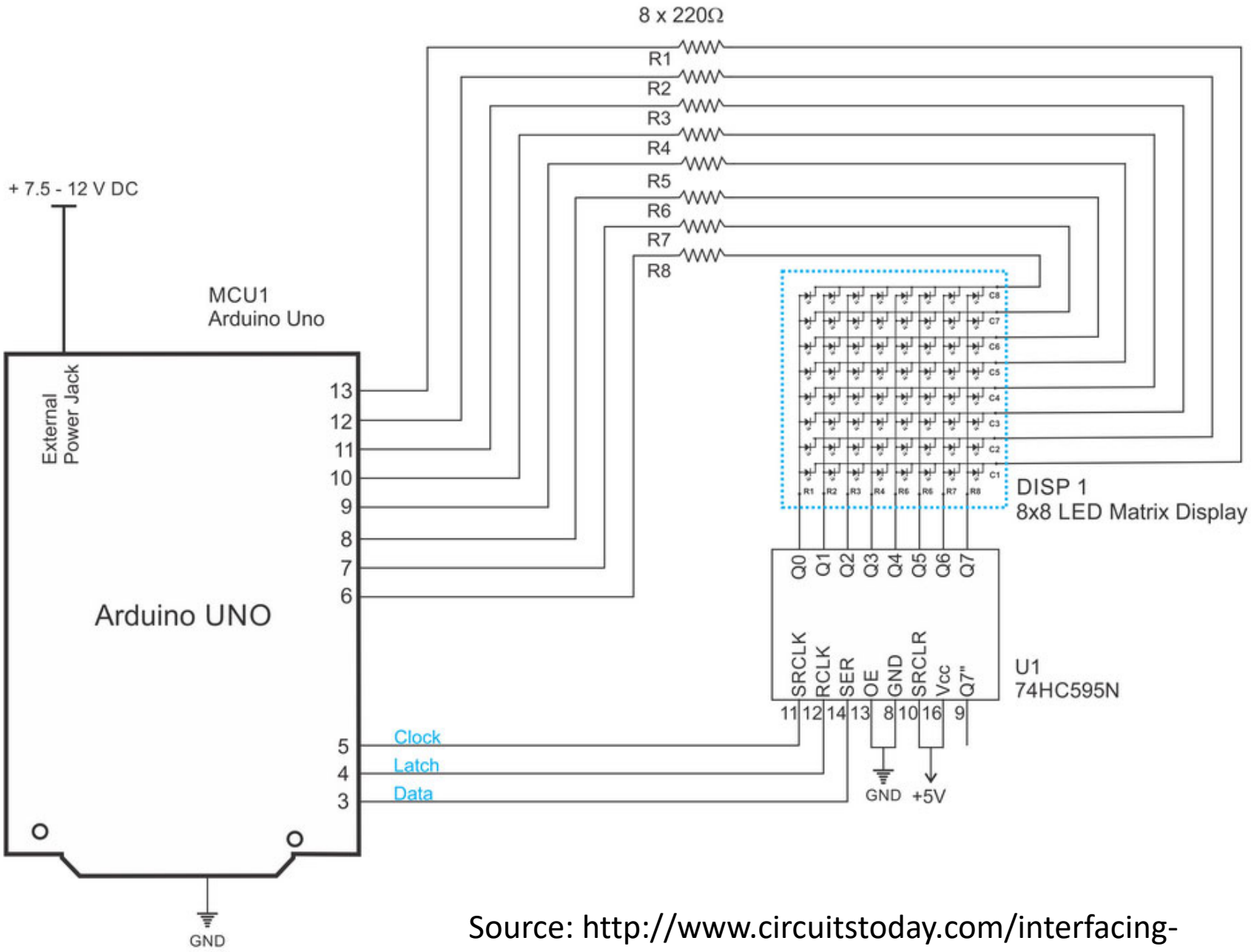
Displaying ‘A’ on 5x7 matrix of LEDs

R1	R2	R3	R4	R5	R6	R7	C1	C2	C3	C4	C5
0	1	1	1	1	1	1	0	1	1	1	1
1 milli second delay											
1	0	0	0	1	0	0	1	0	1	1	1
1 milli second delay											
1	0	0	0	1	0	0	1	1	0	1	1
1 milli second delay											
1	0	0	0	1	0	0	1	1	1	0	1
1 milli second delay											
0	1	1	1	1	1	1	1	1	1	1	0
1 milli second delay											



Source: http://www.circuitstoday.com/interfacing-dot-matrix-led-display-to-8051

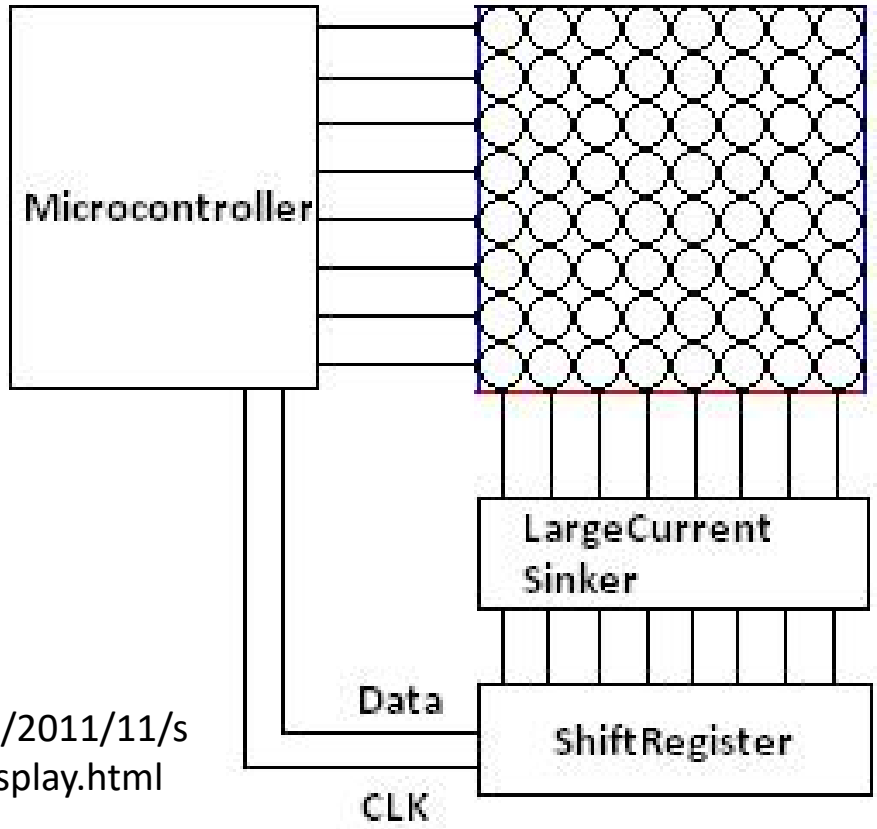
Interfacing to Dot Matrix LED



Source: http://www.circuitstoday.com/interfacing-8x8-led-matrix-with-arduino

Interfacing with Dot Matrix LED

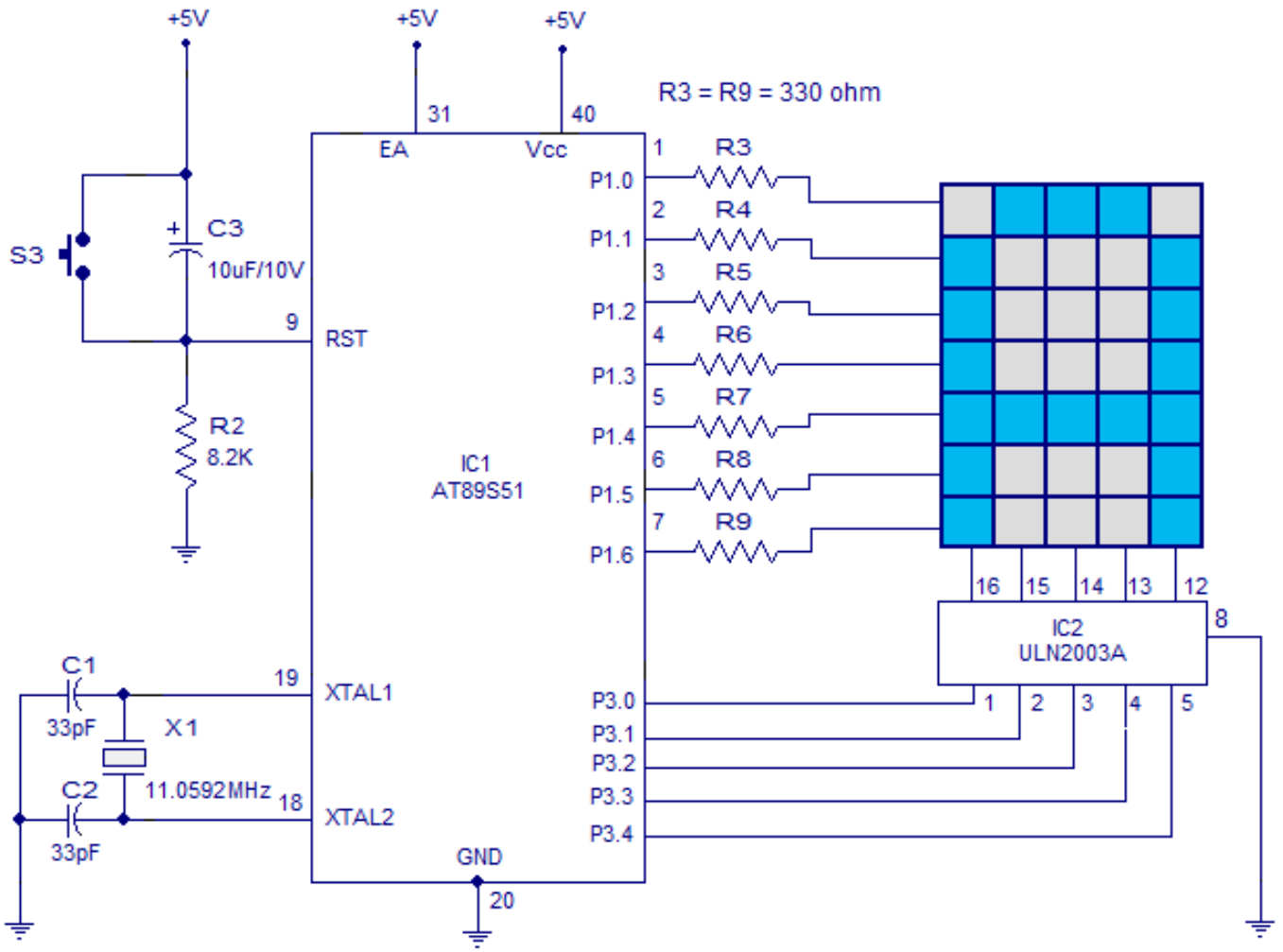
- Across each row, one pin is sourcing the current for only one LED at a time.
- A column pin may have to sink the currents for more than one LED.
- For example, the column C1 must be able to sink the currents from 6 LEDs.
- A microprocessor's I/O pin cannot sink this much of current, so external transistor arrays may be required



Source:
http://www.npeducations.com/2011/11/simple-led-message-scrolling-display.html

Interfacing with Dot Matrix LED

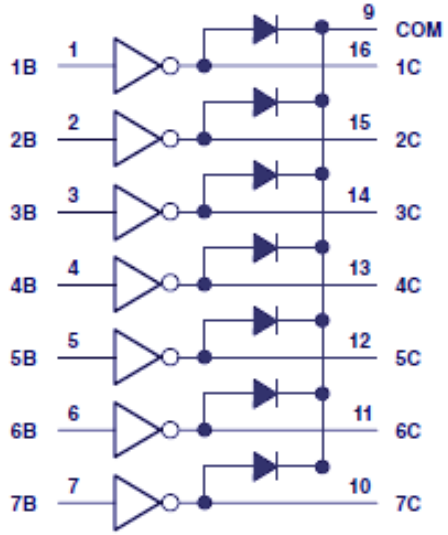
- Current sinking IC such as ULN2003A can be used to drive column lines of array.



Interfacing 7 segment display to 8051

www.circuitstoday.com

Logic diagram

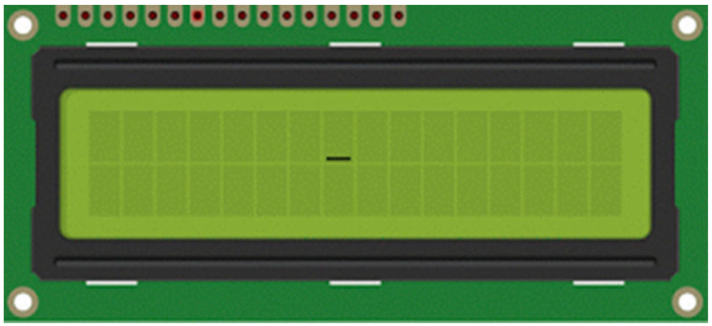


Liquid Crystal Display (LCD)

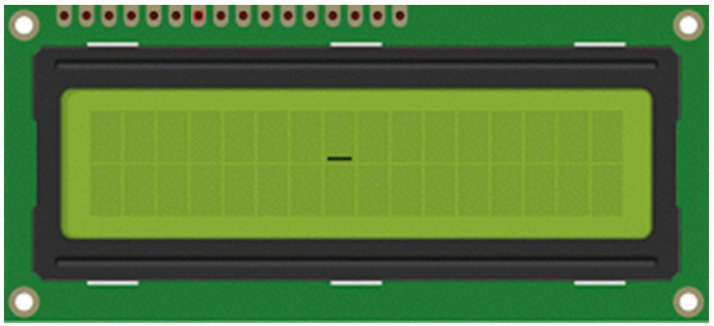
Liquid Crystal Display (LCD)

- Display text, numeric data, graphics
- Easy to interface
- Cheap
- Well documented

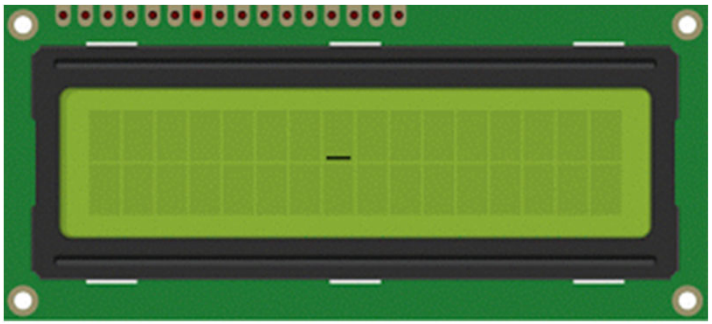
Source: http://embeddedsystemengineering.blogspot.sg/2015/09/lcd-16x2-hd44780-working-principle-part_20.html



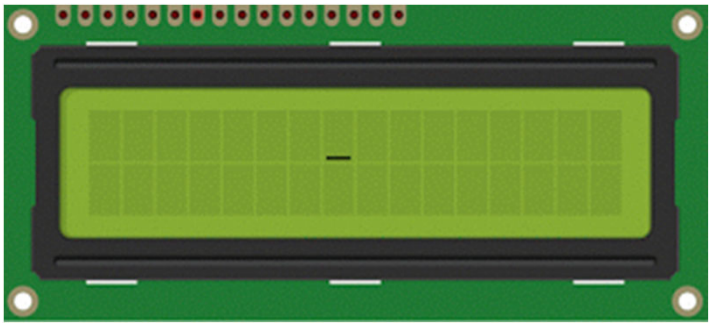
Every new character is entered, Cursor is auto-incremented after each character is entered.



Every new character is entered, Cursor is decremented after each character is entered.



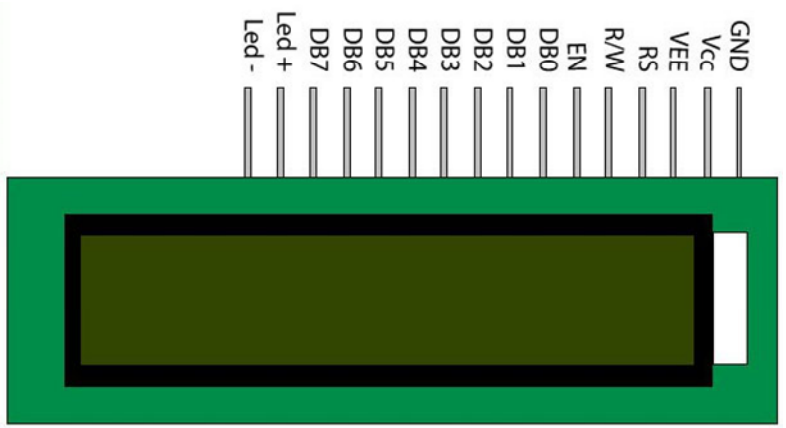
Cursor is auto-incremented and at the same time the display character also auto-shifted to the left.



Cursor is auto-decremented and at the same time the display character also auto-shifted to the right.

LCD Pins

- Most commonly used are 1 Line, 2 Line or 4 Line LCDs.
- They make use of the HD44780 controllers.
- Most of the them 14 pins or 16 pins (2 extra pins are for back-light LED connections).

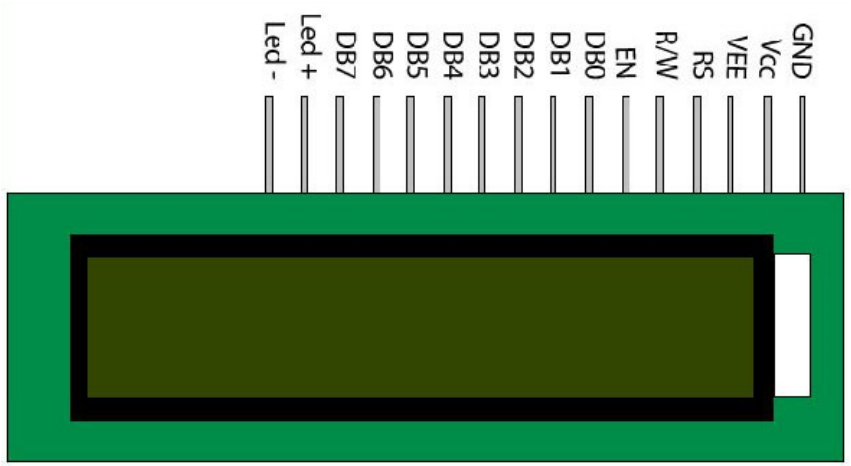


	PIN No.	Name	Function
Power	1	VSS	Ground
	2	VCC	Supply Voltage for Logic
	3	VEE	Supply Voltage for LCD Contrast
Control	4	RS	Register Select
	5	R/W	Read/Write
	6	E	Chip Enable
Data	7	D0	Data Bit 0
	8	D1	Data Bit 1
	9	D2	Data Bit 2
	10	D3	Data Bit 3
	11	D4	Data Bit 4
	12	D5	Data Bit 5
	13	D6	Data Bit 6
	14	D7	Data Bit 7
Backlight	15	A (LED+)	Anode for LED Backlight
	16	K (LED-)	Cathode for LED Backlight

Source: http://binaryupdates.com/interface-lcd-with-lpc2148-arm7/

4-bit vs 8-bit Mode

- 4-bit mode uses 4 data lines and requires two 4-bit transfers for each instruction and character that is sent to the display.
- 8-bit mode uses 8 data lines and requires only one 8-bit transfer for each instruction and character that is sent to the display.
- Trade-Off:
  - 4-bit data transfers will take twice longer to transfer as 8-bit data transfer, but uses lesser lines.
- If I/O is limited, 4-bit mode might preferred. If I/O is enough and timing is important, then go for 8-bit mode.

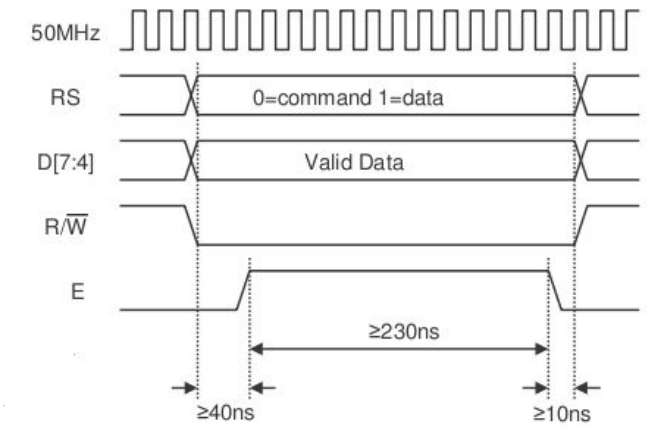


LCD Control Lines

- The 3 control lines are EN, RS and RW
  - EN – Enable
    - This control line is used to tell the LCD that you are sending it data.
    - Must be pulled LOW before any other line is activated.
  - RS – Register Select
    - When RS is LOW, it is treated as a command
    - When RS is HIGH, data is treated as text
  - RW – Read/Write
    - When RW is LOW, information is written to the LCD
    - When RW is HIGH, the program is querying the LCD.
    - Only 1 instruction, “Get LCD Status” is a Read command.

LCD Commands

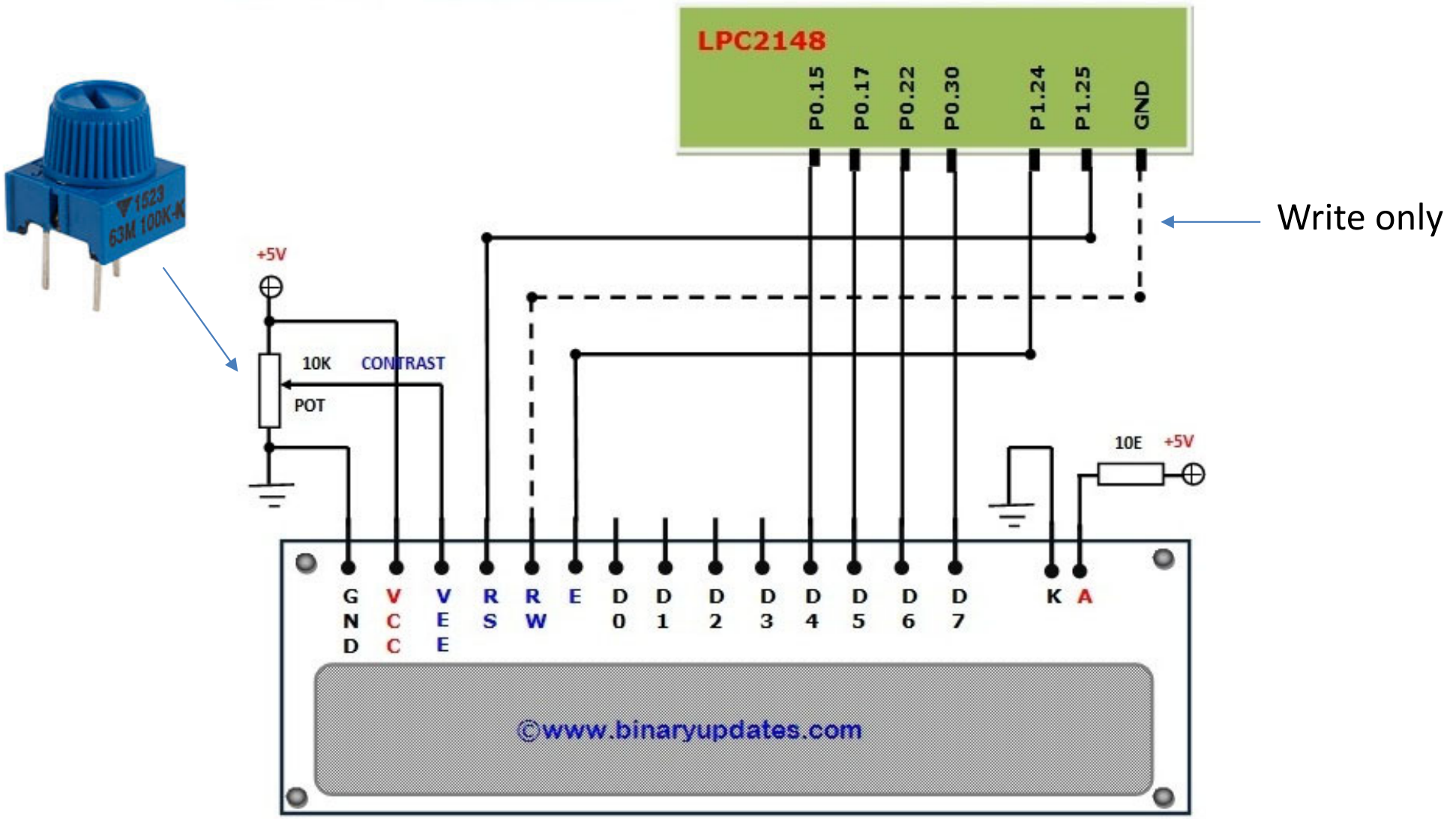
- To establish proper communication and interface between LCD with microcontroller, we need to supply commands in a given order to the data pins to initialize LCD properly.
- The 3 control lines and data lines are used to issue the commands



Step: 1 Put MSB of the data
Step: 2 EN = 0; RS=0
Step: 3 EN=1 wait EN=0 (**Clock** to latch MSB)
Step: 4 put LSB of the data
Step: 5 EN = 0; RS=0
Step: 6 EN=1 wait EN=0 (**Clock** to latch LSB)
Step: 7 Wait

No	HEX Value	COMMAND TO LCD
1	0x01	Clear Display Screen
2	0x30	Function Set: 8-bit, 1 Line, 5x7 Dots
3	0x38	Function Set: 8-bit, 2 Line, 5x7 Dots
4	0x20	Function Set: 4-bit, 1 Line, 5x7 Dots
5	0x28	Function Set: 4-bit, 2 Line, 5x7 Dots
6	0x06	Entry Mode
7	0x08	Display off, Cursor off
8	0x0E	Display on, Cursor on
9	0x0C	Display on, Cursor off
10	0x0F	Display on, Cursor blinking
11	0x18	Shift entire display left
12	0x1C	Shift entire display right
13	0x10	Move cursor left by one character
14	0x14	Move cursor right by one character
15	0x80	Force cursor to beginning of 1st row
16	0xC0	Force cursor to beginning of 2nd row

Interfacing with LCD using 4-bit Mode



Source: http://binaryupdates.com/interface-lcd-with-lpc2148-arm7/

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Copyright Notice

All course materials (including and not limited to lecture slides, handouts, recordings, assessments and assignments), are solely for your own educational purposes at NTU only. All course materials are protected by copyright, trademarks or other rights.

All rights, title and interest in the course materials are owned by, licensed to or controlled by the University, unless otherwise expressly stated. The course materials shall not be uploaded, reproduced, distributed, republished or transmitted in any form or by any means, in whole or in part, without written approval from the University.

You are also not allowed to take any photograph, video recording, audio recording or other means of capturing images and/or voice of any of the course materials (including and not limited to lectures, tutorials, seminars and workshops) and reproduce, distribute and/or transmit in any form or by any means, in whole or in part, without written permission from the University.

Appropriate action(s) will be taken against you (including and not limited to disciplinary proceedings and/or legal action) if you are found to have committed any of the above or infringed copyright.

1 August 2022